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Multi-objective evacuation routing in transportation networks

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A R T I C L E I N F O

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A B S T R A C T

In this paper, the optimal design and analysis of evacuation routes in transportation networks is examined. An methodology for optimal egress route assignment is suggested. An integer programming (*IP*) formulation for optimal route assignment is presented, which utilizes $M/G/c/c$ state dependent queueing models to cope with congestion and time delays on road links. $M/G/c/c$ simulation software is used to evaluate performance measures of the evacuation plan: clearance time, total travelled distance and blocking probabilities. Extensive experimental results are included.

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1. Introduction

Natural and man-made emergency events, such as hurricanes, floods, wildfires or chemical spills, can impose serious risk and threat on the well-being and safety of a population. In this case, massive evacuation or in site sheltering (shelter-in-place) has been used as means of protecting a population from potential harm (Sorensen et al., 2004). To shelter-in-place is a preferred option if there is a risk for a population to be exposed to some chemical agents or hazardous materials and if shelter provides adequate level of protection. However, evacuation is a procedure most often used in case some infrastructure or community can be damaged by hurricanes, floods or wildfires (Perry and Lindell, 2003). Scientific or formal analysis of the evacuation processes is very complex due to the large number of evacuees and unforeseen loads on the transportation infrastructure. Such analysis of emergency situations can require synthesis of knowledge from many different areas of expertise: social and behavior science, transportation, health care, telecommunication, security, police, forestry, and so on.

1.1. Objectives of research

The purpose of this paper is to design and to analyze egress route assignment algorithms for regional evacuation in large transportation networks. This problem is challenging due to the fact that the speed of vehicles decreases with an increase in the number of vehicles on the road segments as well as of the potential blocking on heavy utilized road segments. Stochastic arrivals of evacuees from affected areas and peak demands for transportation infrastructure add much complexity to the problem. An integer programming (*IP*) model is suggested to compute an optimal routing policy for the population from the effected areas, where a set of feasible and potential egress routes is defined with the kth shortest path algorithm. An embedded $M/G/c/c$ state dependent queuing model is used to evaluate the travel time along the road links. Finally, a state-of-the-art simulation model *MGCCSimul* is utilized to evaluate the resulting clearance time, travel distance, and level of congestion for the designed evacuation plan. We illustrate the suggested approach with a case study and extensive experimental results.

1.2. Outline of paper

Section 2 provides a review of current research in the area of evacuation planning and modeling. Section 3 of the paper describes the overall problem. Section 4 outlines the overall framework for the methodology and Section 5 discusses the solution methodology and algorithm. Section 6 describes the case study and the experimental results and Section 7 concludes the paper.

2. Literature review

In the literature, relocation from areas at risk to areas of greater safety is referred to as an *evacuation* (Southworth, 1991; Zelinsky and Kosinski, 1991). Regarding its properties, evacuation may differ by scale, objects of relocation (people vs. property), and by level of control by authorities. In addition, the evacuation can be mandatory, recommended, or voluntary and should be conducted according to an evacuation plan. Evacuation is a complex process consisting of several consecutive phases (Fig. 1). The first phase

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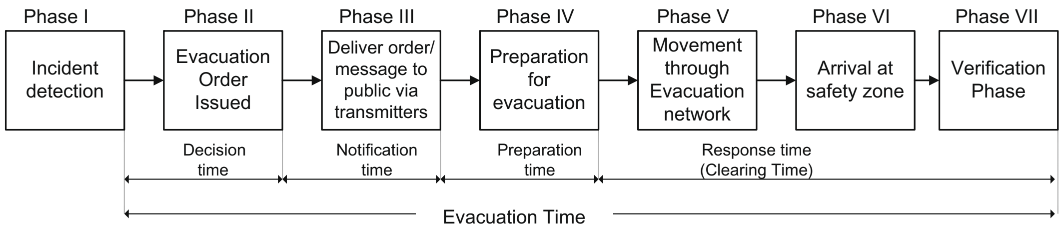

Fig. 1. Evacuation phases.

is detection of an incident. In Phase II, decision makers have to evaluate the risk and potential threat for specific areas. An evacuation order should be issued for these areas if the risk is significant and there are no shelters to provide adequate in-place protection (Lindell, 1995). (These areas constitute origins of evacuation or evacuation planning zones (*EPZ*)). Phase III, the alert, has to be communicated to the population. During the next phase, the population makes a decision to evacuate or not to evacuate depending on their perception of danger. This phase also implies preparation for leaving. Phase V implies movement of population through a transportation network to designated safe areas (destinations) or centers of refuge. This step involves clearing of occupants from affected areas. Finally, in Phase VI, evacuees arrive to areas outside of danger (destinations) and a verification that all evacuees have made it safely must be carried out, which is Phase VII. It is important to note that time intervals for Phases III–VI represent average time for all groups of evacuees, as these steps may have different durations for each class of evacuees.

Depending on the scale of a emergency situation, neighborhoods, towns or regions may have to be evacuated. The evacuation time may range from hours to weeks or even months (Church and Sexton, 2002). Generally, capacity of transportation networks cannot satisfy the intense demand for transportation during the evacuation. Even for small neighborhood-scale evacuations, transportation networks impede the fast clearing of the population from an effected area (Church and Sexton, 2002; Cova and Johnson, 2002). The magnitude of the problem is greater for regional evacuations. To manage such emergencies effectively, the decision makers may benefit from having in-place evacuation plans (*EP*). The aim of an evacuation plan is to define optimal evacuation policies for the population from areas under risk and uncertainty. Decision makers design and update evacuation plans for scenarios which are most likely to happen. This approach is known as preplanning, though real-time design or re-evaluation of evacuation plans are required in cases of infrastructure failure, road blocks, and unexpected changes in direction and speed of emergency event propagation (Alexander, 2002).

2.0.1. Emergency evacuation planning

As we pointed out in the previous section, controlled evacuation is governed by a specific evacuation plan, which defines the actions of the population and authorities. This action plan is a result of a purposeful emergency evacuation planning (*EEP*) process. An emergency evacuation planning is considered as a five-step process (Southworth, 1991). If the evacuation planning process occurs before Phase I of the evacuation (Fig. 1), then it's known as preplanning. In this case, an action plan is designed *prior* to an emergency event for a finite set of potential scenarios. If it occurs during Phases II–V, then the evacuation planning process can be classified as *real-time*. We will discuss these *EEP* steps in more detail in the following sections.

The fact that many plans are designed *prior* to an evacuation and for a *limited* number of scenarios is defined by the complexity of the regional evacuation process, limited resources and lack of well-defined protocols. In practice, assumptions made during such planning can be made invalid during an emergency due to changes in weather conditions or in an emergency event behavior, and destruction of transportation infrastructure. Real-time emergency evacuation planning requires that an integrated methodology be available for decision makers. Such a methodology combines efficient analytical methods for modeling emergency event behavior, optimal routing assignment, and allows one to manage the evacuation process in real-time.

2.0.2. Step 1: traffic generation

Evacuation planning zones (*EPZ*) or areas, predicted to be affected by an emergency event, act as origins of evacuee traffic. The ultimate purpose of this step is to define a number of vehicles leaving the areas and loading an evacuation network. First, it is necessary to evaluate the population size presented in a specific area at a particular time. Then, this population size should be translated into the size of the vehicle fleet used for evacuation. Population distribution differs significantly between day and night time, as people commute to work, schools, shopping and service centers. Major data providers (the US Census program) have accurate information on night time population distribution, in other words this is information on where people live. An evaluation of population size during daytime is a challenging issue (Southworth, 1991). Daytime population comprises population at work, at school, at service centers (hospitals, medical care facilities, sport gyms) and population, which stay at home (retirees, small children, etc.). In addition, there are special groups of population with limited mobility (hospitals, nursing homes, correctional/penitary facilities). These population groups are transit dependent. Finally, through traffic may generate additional loads on the evacuation network. Lindell and Prater (2007) provide detailed discussions of trip generation techniques based on empirical behavioral studies.

2.0.3. Step 2: evaluation of traffic loading rate

The purpose of this step is to evaluate the time distribution of the evacuees' departure process. Southworth (1991) outlined four major approaches to define traffic loading curves, which represent the fraction of evacuated population at a specific time. The assumptions about evacuees' behavior are based on empirical data and historical reviews, surveys of intentions and expert judgment, simulations of alarm message propagation and perception in the community (Southworth, 1991; Cohn et al., 2006; Stern and Sinuany-Stern, 1989). Lindell and Prater (2007) provide a comprehensive overview of factors affecting departure time distribution, analysis of distribution used in practice, as well as comparison with empirical data.

Valid assumptions on evacuees behavior for a particular type of hazardous events is crucial for accurate evacuation modeling. Evacuees behavior is a complex phenomenon, for instance, many people don't leave the region after being warned (Sorensen et al., 2004). In some situations, on the contrary, a significant part of the population have decided or already have left prior to the moment when the evacuation was officially declared (Lindell et al., 2005). Cohn et al. (2006) provide some insight on population behavior using post surveys of three major wildfire evacuations. Using empirical data, Lindell et al. (2005) analyze households response during hurricane Lili. The decision to leave depends on many factors: person's perception of danger, family and social status, existence of family/friends at the destination points, or even type of emergency event (Stern and Sinuany-Stern, 1989). People tend to take floods and chemical spills more seriously, than hurricanes and wildfires. Some people decided to stay to protect their property from fire or because of pets.

In practice, the timing response is described with traffic loading *S*-shaped curves. For instance, Eq. (2), suggested by Radwan and modified by Southworth and Chin, explains the cumulative percentage of total traffic P_t loaded into evacuation network at time t (Radwan et al., 1985; Southworth and Chin, 1987; Southworth, 1991; Radwan, 2005)

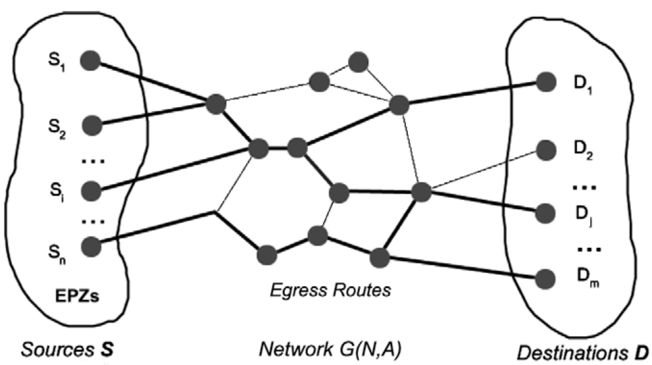

Fig. 2. Evacuation network $G_e(N,A)$ connects *EPZs* and destinations.

$$P_t = \frac{1}{1 + d \cdot \exp(at)}, \tag{1}$$

$$d = \frac{1 - P_0}{P_0}, \tag{2}$$

where a is a parameter to calibrate and d is ratio of vehicles to be evacuated to vehicles which are already in the road network at time $t = 0$. There are reported applications of the Rayleigh distributions (Tweedie et al., 1986) and exponential distributions (Hobeika and Kim, 1998) for estimating time evacuees departure time. In their model, Cova and Johnson (2002) view evacuees' departure as a Poisson process.

2.0.4. Step 3: definition of refuge centers

At this step of the evacuation planning process, the destinations should be identified. Lindell and Prater (2007) point out that for evacuation planning purposes both evacuees' ultimate and proximate destinations should be considered. Ultimate destinations are public facilities and private establishments where evacuees *stay* before returning back home. The majority of evacuees stay with relatives, friends or in hotels. In case of public refuge centers, they should provide enough space to accommodate evacuees and be located in areas with adequate ingress routes to deliver humanitarian aid, medical assistance, etc. From a practical point of view, sport arenas, public schools or universities buildings may be used as they were designed to hold large numbers of people. In turn, proximate destinations are viewed as boundaries, after crossing, in which evacuees are safe. For more discussion on destination choice, please refer to Lindell and Prater (2007).

2.0.5. Step 4: traffic route assignment

At this step of the evacuation planning, we assume that *EPZs*, destinations and evacuation transportation networks have been defined. Demand and load patterns of evacuee's have been evaluated as well. Then, the purpose of this step is to determine optimal routes and assign vehicle fleets to them (Fig. 2). This is a complex

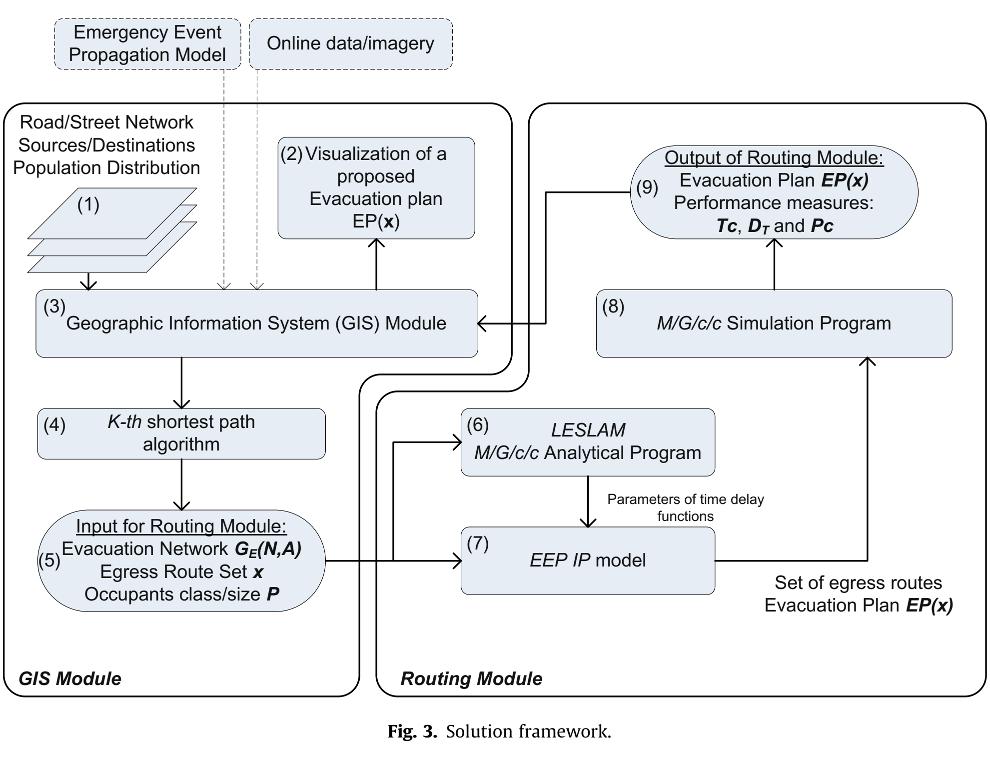

Fig. 3. Solution framework.

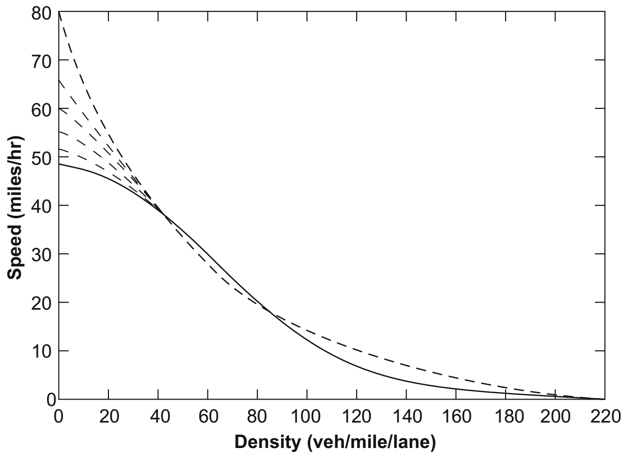

Fig. 4. Speed-density curve: road link performance decays with increase in traffic density (Source: (Smith, 2001)).

optimization problem, which has *NP*-hard complexity (Kerbache and Smith, 2000). One of the inherited properties of the problem is that the evacuation network performance depends on traffic load. Velocity of vehicles on particular road links decreases with increasing of number of vehicles (Fig. 4). Over-utilization of a particular road link can block traffic through the link completely. Such blocking will affect all upstream links and can cause significant congestion through the evacuation network (Cruz and Smith, 2005; Cruz et al., 2005; Talebi and Smith, 1985).

The design of an optimal evacuation policy comprise the following three major sub-problems. First, it is necessary to define the functional criteria of an evacuation plan and suggest quantitative measures of the evacuation. The second is to develop routing policies (as a set of egress routes). Finally, evaluation of evacuation plan's effectiveness should be performed. In the practice of evacuation modeling, total clearance time, total traveled distance, queue size and blocking probability are the most common criteria which should be extremized.

2.0.6. Step 5: verification of an evacuation plan

This step in the evacuation planning procedure concludes the design of an evacuation plan with its performance evaluation. If the designed plan is prone to traffic congestion or fails to provide timely evacuation, some alternative scenarios or application of traffic control tools should be considered. Decision makers may consider reversing direction of traffic flow on some road links and using contraflow lanes (Wolshon, 2001; Wolshon and Lambert, 2006), disabling ramps and exists, changing routing at intersections to minimize traffic conflicts (Cova and Johnson, 2003). Then *Step* 4 of the procedure should be repeated.

This paper focuses mainly on *Step* 4 "traffic route assignment" and uses information from the previous steps as input for analysis, as we assume that dynamics of emergency event propagation, origins, population sizes and departure characteristics, and transportation network were defined.

2.0.7. Evacuation models review

Historically, evacuation modeling approaches were influenced greatly by fields of operations research (*OR*) and transportation engineering (*TE*). *OR* approaches aim to minimize total evacuation time for the whole system or for individual user. System optimization methods consider traffic as non-interrupted flows, which satisfy demand existing in destination nodes, and apply different network optimization models to allocate flows optimally without exceeding evacuation network capacity. Methods, which apply user equilibrium (*UE*) techniques, minimize travel time for each vehicle (Algers et al., 1997; Sheffi et al., 1982; Hobeika and Kim, 1998).

Evacuation models, which represent traffic as flows, belong to macro-level models. Transportation engineering approaches often consider traffic at more detailed levels of an individual vehicle or a group ("platoon") of vehicles. Such approaches can be classified as "micro" and "meso" models. The evacuation models can be classified also regarding their route assignment procedure. Static assignment implies that vehicles stay on a specific route during the evacuation, which can be a realistic feature of mandatory evacuation with imposed route controls (roads closure, police regulating traffic). Other models consider dynamic assignment, where a vehicle chooses a future direction at every road crossing. Modeling techniques can be grouped by computational techniques into analytical and simulation techniques. Table 1 summarizes characteristics of several evacuation models.

Table 1
Classification of evacuation models

	Model	Type	Route assignment type	Computational techniques	Authors (year)
1	Clear	Micro	Simple		Moeller et al. (1982)
2	DYNEV I-DYNEV	Meso	Static	Simulation with embedded *UE* algorithm	KLD Associates (1984)
3	DYMOD	Macro	Dynamic	Simulation with embedded optimization routine	Southworth et al. (1992)
4	MASSVAC4	Macro	Dynamic	Simulation with embedded *UE* and Dial's algorithm	Hobeika and Kim (1998)
5	NETVAC1	Macro	Dynamic		Sheffi et al. (1982)
6	TRANSIMS	Micro	Dynamic	Parallel simulation	Pickert and Nagel (2001)
7	CEMPS	Micro	Dynamic	C++ discrete event simulation coupled with GIS	Pidd et al. (1996)
8	$M/G/c/c$ state dependent queueing model	Micro	Static	Analytical technique	Jain and Smith (1997)
9	MGCCSimul	Micro	Dynamic	Simulation with $M/G/c/c$ state-dependent queues. Support modeling of vehicle and pedestrian traffic	Cruz et al. (2005)
10	OREMS (by ORNL)	Micro & dynamic	Static		Rathi and Solanki (1993), Bhaduri et al. (2006), Franzese and Sorensen (2001)
11	DYNASMART	Micro Micro	Dynamic		Mahmassani et al. (1995)
12	EMBLEM2	Micro	Dynamic	Empirical behavioral model	Lindell (2008)
13	A lane-based evacuation routing model	Macro	Static	Embedded OR procedure to control intersections	Cova and Johnson (2003)

From a review of the evacuation models (Southworth, 1991; Algers et al., 1997; Smith, 2001; Santos and Aguirre, 2004), two approaches are evident. The first approach defines a set of optimal routes and evaluates performance measures simultaneously. The second approach uses an analytical optimization technique to offer a routing policy, and then this policy is evaluated with a traffic simulation model (Talebi and Smith, 1985; Smith, 1991, 1994). According to model reviews, the first approach is prevalent in practice. However, the second approach is overlooked and should be examined in more detail.

3. Problem formulation

This problem is situated within the domain of *optimal routing problems (ORTP)* (Kerbache and Smith, 2000; Smith, 2001). The class of *ORTP* considers the issues of routing of evacuees through an existing transportation network and addresses potential congestion and delays which could be created with the routing decision. This problem is known to be *NP*-Hard (Kerbache and Smith, 2000; Smith, 2001), due its multi-commodity nature, as evacuees can be differentiated by their origin–destination pairs as well as by their class or group. During an evacuation, we need to safely guide and relocate the occupants (population) P_i from the affected areas (sources) S_i to destinations D_j through a transportation network $G(N,A)$ over time t. Each population class/group P_i should be assigned to an egress route (Fig. 2). The objective here is to find a set of egress routes to *safely* evacuate the occupant population P over G. Further we will refer to a set of potential routes as to an evacuation plan (*EP*), which we denote with the following notation: $EP = \{R_{S_1}, R_{S_2}, \ldots, R_{S_n}\}$, where R_{S_i} is an egress route from source S_i.

This problem is complex due to its stochastic, integer programming, and multi-criteria nature. As occupants attempt to use shortest routes some of the traffic will over utilize various road segments, causing congestion and potential blocking along those routes. One of the ways to attain population *safety* during the evacuation process is minimization of total clearance time T_C, total traveled distance D_T and congestion on road links (Smith, 2001; Kerbache and Smith, 1987, 1984). Therefore, the method described in this article allows one to generate an evacuation plan $EP(x)$ (a vector of egress routes $\mathbf{x}$ for population P) that minimizes simultaneously total travelled distance (D_T), blocking (P_C) and total clearance time (T_C) along the egress routes.

3.1. Notation

In this section, we propose an integer set-packing formulation of the stochastic *EEP* problem. The notation used in the problem formulation is introduced below:

Indexes

- i — an index for population sources (origins) S_i;
- j — an index for destinations D_j;
- k — an index for kth shortest paths;
- l — an index for a set of road links A;

Decision variables

- x_{ijk} — 1 if population was assigned to kth shortest egress route connecting source i and destination j (otherwise $x_{ijk} = 0$)

Parameters

- w — a weight coefficient $w \in [0, 1]$ and should be set by a decision maker.
- D_T — total distance traveled by occupants
- T_C — clearance time for the system;
- $D_T^{\min}$ — total distance when all occupants are assigned to the 1*st* shortest routes
- $T_C^{\min}$ — the best clearance time of the system (in absence of congestion) and all occupants travel along shortest egress routes with maximal velocity
- d_{ijk} — total length of route x_{ijk}
- P_i — total number of occupants from source i;
- λ_i — an arrival input rate into the system from source i
- Λ — a vector of arrival rates $(\lambda_1, \lambda_2, \ldots)$
- λ_l^L — a total arrival rate into road link l
- $\lambda_l^{L\max}$ — a maximum arrival rate into road link l such that $P_C(\lambda_l^{L\max}) \leqslant \epsilon$, where ϵ is a threshold value of blocking probability
- $t_l^{(1)}$ — minimum expected traverse time through a road link l
- $t_l(\lambda_l^L)$ — *expected* traverse time through a road link l given an arrival rate λ_l^L
- $V_l^{(n)}$ — a maximum velocity of occupants on road link l, where n occupants present on the road link
- L_l — distance (length) of road link l
- α_{likj} — *1* if egress route x_{ijk} includes road link l (0 otherwise)
- C_j — capacity of destination refuge center D_j

3.2. Mathematical formulation

The multi-objective model to solve the *EEP* problem is suggested below:

$$\min_{\mathbf{x}} Z = w\frac{1}{D_T^{\min}}\sum_i P_i \sum_j \sum_k x_{ijk} d_{ijk} + (1-w)\frac{1}{T_C^{\min}}\sum_l t_l(\lambda_l^L) \times \sum_i P_i \sum_j \sum_k x_{ijk}\alpha_{lijk} \tag{3}$$

subject to:

$$\sum_j \sum_k x_{ijk} = 1 \text{ for } \forall i \quad (\text{Sources}), \tag{4}$$

$$\sum_i \sum_k P_i x_{ijk} \leqslant C_j \text{ for } \forall j \quad (\text{Shelters}), \tag{5}$$

$$\sum_i \lambda_i \sum_j \sum_k \alpha_{lijk} x_{ijk} \leqslant \lambda_l^{L\max} \text{ for } \forall l \quad (\text{Roadlinks}), \tag{6}$$

$$x_{ijk} = 0 \text{ or } 1 \text{ for } \forall i \text{ and } \forall k \quad (\text{Routes}). \tag{7}$$

The objective function Z is a linear combination of excess total travel distance D_T and excess clearance time T_C. The *excess* travel distance evaluates deviation of total traveled distance D_T (which is the performance measure of a routing assignment $\mathbf{x}$) from the best or minimal travel distance $D_T^{\min}$

$$D_T^{\min} = \sum_{i=1}^{n} P_i \sum_{j=1}^{m} x_{ij1} d_{ij1}. \tag{8}$$

The occupant population will travel $D_T^{\min}$ miles if and only if the evacuees are assigned the 1st shortest egress routes, therefore the value of $D_T^{\min}$ is the lower bound for any evacuation plan $EP(\mathbf{x})$ (Eq. (8)). Eq. (9) represents the minimum clearance time $T_C^{\min}$, which defines the lower bound on clearance time T_C for any egress routing $\mathbf{x}$ and corresponds to the best case scenario, when population is evacuated along 1st shortest egress routes with maximum velocity $V_l^{(1)}$ at any road link l

$$T_C^{\min} = \sum_l t_l^{(1)} \sum_{i=1}^{n} P_i \sum_{j=1}^{m} x_{ij1}\alpha_{lij1}. \tag{9}$$

This above problem is difficult to solve since

- an optimal solution is a *trade-off* among three specified criteria: congestion, clearance time and total travelled distance;

- the parameters of the model t_l and P_C depend upon the stochastic model which evaluates a given set of routes.

In the above model, Eq. (6) defines a constraint, which captures the traffic congestion minimization. The probability of blocking P_C on a road segment a_l depends on evacuees arrival intensity λ_l^L. We discuss an approach to compute an upper bound on input arrivals $\lambda_l^{L_{\max}}$ which will assure the blocking probability will not exceed the threshold value ϵ, therefore $P_C(\lambda_l^{L_{\max}}) \leqslant \epsilon$. By setting a threshold value ϵ on the maximum blocking probability in the evacuation network $G_e(N,A)$, we assure that evacuees' flows on egress routes are non-interrupted.

Constraint set, defined by Eq. (4), ensures that all population from a source node i is evacuated along one egress route. This formulation allows one to model multiple class evacuation with introduction of "dummy" source nodes with specific population size and arrival rates. Eq. (5) represents the constraint set which captures aggregate capacity restrictions for destination D_j. We assume that destination node D_j is an ultimate destination for some part of evacuees and a proximate destination for the rest of people. Therefore, C_j is an upper limit on total number vehicles which will arrive at D_j and stay there at shelters, hotels or at relatives, and vehicles, which will go through D_j to other areas.

The suggested model formulation has two unique features, which require thorough explanation. The first feature ensures the non-interrupted movement of evacuees via the evacuation network and absence of blocking on any road segment. The second feature copes with time delay modeling along egress routes given that vehicles velocity is decreasing function of number of vehicles at the road segment and that arrivals as well as number of vehicles on the road segment is *random* value. Both features are based on a $M/G/c/c$ state-dependent decay service queueing model to model traffic congestion on a road link, which is a core of the proposed IP formulation.

The next sections will describe treatment of time delay functions $t_l(\lambda)$ and congestion level in detail.

4. Solution framework

The exploration of vehicle routing problems in transportation networks is performed with the following framework, see Fig. 3. This system is based on modules developed in the *Dynamic Facilities Layout and Simulation Modeling Laboratory*, UMass. The system includes three major components: (i) Geographic Information System (GIS) component (ii) Routing (evacuation) planning component and (iii) a set of utilities, which combine two components together. The GIS component is based on ArcView GIS 9.2©. The GIS module (block 3 in Fig. 3) is responsible for entering, storing, querying and representing spatial information. The following information is used for evacuation planning: population location and characteristics, origin–destination data, road network topology, road links characteristics, etc. The kth shortest path algorithm (JimTnez and Marzal, 2003; Eppstein, 1998) is utilized to compute an egress route set for each origin–destination pair. Therefore the road links, which comprise egress routes, define an evacuation network $G_E(N,A)$. This information, together with occupants' data, is used as input data for the routing module. The Routing (traffic assignment) module includes three major sub-modules. The first one is a *LESLAM* module, which uses road link data to define time delay functions $t(\lambda)$ for each road segment. The second, *EEP IP* model, solves the route assignment problem and finds a Pareto optimal evacuation plan (set of evacuation routes) for specified scenario. The third module is a state-of-the-art vehicular evacuation simulation model (Cruz et al., 2005), which is used to evaluate evacuation plan's performance. The following measures of effectiveness are computed: estimated clearance time T_C, and total travelled distance D_T, as well as detail statistics for each road link, such as: congestion level (probability of blocking P_C), throughput θ, average number of vehicles on the road segment $E(Q)$ and average traverse time T_S. After that the resulting solution is visualized with the GIS module and an evacuation planner (a decision maker) can obtain an evacuation plan map along with evacuation plan performance characteristics. GIS has been an important tool for emergency management and planning (de Silva and Eglese, 2000; Church and Sexton, 2002; Pal et al., 2003; Zerger and Smith, 2003), which provides adequate and effective ways to represent and manage transportation network models, as well as to integrate such models with required socio-demographic data for the modeled area. The road networks of different scale and detail level (from major highways to streets) and demographic data can be obtained from the Department of Transportation and from State cartographic agencies as well as from private data providers. In addition, an ability of GIS to include high-resolution imagery and online satellite data, aids a decision maker in preparing a realistic network model, as some specifics of road configurations, topology, available ramps or exists can be verified.

5. Solution methodology

We follow an approach suggested by Smith et al. (Smith, 1991; Smith, 1994; Karbowicz and Smith, 1984; Cruz et al., 2005; Talebi and Smith, 1985) to model traffic and congestion on a single road link and Markovian arrivals into the road link. This $M/G/c/c$ queueing model allows one to capture traffic congestion and its stochastic nature on a road link with specific physical characteristics as length L, number of lanes W and maximum density $K_{\max}$ (Fig. 5) and total capacity of $c = K_{\max}LW$.

The queueing model updates service rate with changes in number of vehicles in the system (Fig. 4) and allows one to evaluate system in term of expected service time, expected number of vehicles in the system, probability of n-vehicles being in a road link, etc. Therefore, a road segment can be modeled as a queueing service facility with the following physical characteristics:

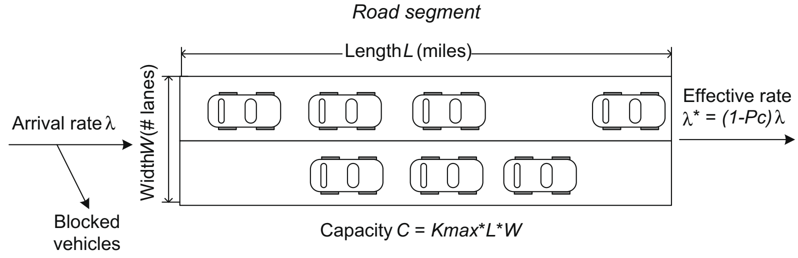

Fig. 5. Representation of a road link as $M/G/c/c$ queueing system.

(i) A queue has a finite size since a road segment has finite length and width(capacity), or number of lanes and can accommodate a limited number of vehicles. Capacity c of a road segment can be computed as $c = K_{max}LW$, where K_{max}, L, W are jam density, length and number of lanes, respectively.
(ii) Vehicles arrive according to Markovian Process.
(iii) Travel time along the road segment is considered as a service time in the queueing facility.
(iv) The queue can be classified as state-dependent because the service rate μ is a function of number of vehicles along the road segment (Fig. 4).
(v) Number of servers is limited with capacity c of the road segment. The queue facility cannot serve more than c vehicles. In other words there is no buffer or waiting space, therefore as soon as a vehicle enters the road link, it starts obtaining the service.
(vi) Service rate distribution is general (G) due to service rate decay.

Table 2
Congestion models

Linear congestion model	Exponential congestion model
$V^{(n)} = \frac{V^{(1)}}{c}(c+1-n)$	$V^{(n)} = V^{(1)} \exp\left[-\left(\frac{n-1}{\beta}\right)^{\gamma}\right]$
	where $\gamma = \ln\left[\frac{\ln(V_a/V^{(1)})}{\ln(V_b/V^{(1)})}\right] \Big/ \ln\left(\frac{a-1}{b-1}\right)$
	$\beta = \frac{a-1}{[\ln(V^{(1)}/V_a)]^{1/\gamma}} = \frac{b-1}{[\ln(V^{(1)}/V_b)]^{1/\gamma}}$
$f(n) = \frac{V^{(n)}}{V^{(1)}} = \left(\frac{c+1-n}{c}\right)$	$f(n) = \frac{V^{(n)}}{V^{(1)}} = \exp\left[-\left(\frac{n-1}{\beta}\right)^{\gamma}\right]$
$P_n = \left[\frac{[\lambda L/V^{(1)}]^n}{\prod_{j=1}^{n} j[(c+1-j)/c]}\right] P_0$	$P_n = \left[\left(\frac{\lambda L}{V^{(1)}}\right)^n \Big/ \prod_{j=1}^{n} j \exp\left[-\left(\frac{j-1}{\beta}\right)^{\gamma}\right]\right] P_0$
where	where
$P_0^{-1} = 1 + \sum_{i=1}^{c} \left[\frac{[\lambda L/V^{(1)}]^i}{\prod_{j=1}^{i} j[(c+1-j)/c]}\right]$	$P_0^{-1} = 1 + \sum_{i=1}^{c} \left[\left(\frac{\lambda L}{V^{(1)}}\right)^i \Big/ \prod_{j=1}^{i} j \exp\left[-\left(\frac{j-1}{\beta}\right)^{\gamma}\right]\right]$

Where c is the capacity of a road link l; n is the number of vehicles with a road link $n \in (0,\ldots,c)$; a is the density of vehicles $K_{dens} = a$; b is the density of vehicles $K_{dens} = b$; V_a is the average travel speed at vehicle density $K_{dens} = a$; V_b is the average travel speed at vehicle density $K_{dens} = b$; $V^{(n)}$ is the average travel speed for n vehicles within a road link; $V^{(1)}$ is the maximum travel speed in free flow conditions (speed limit); β, γ are the scale and shape parameters for exponential congestion model.

The assumptions above allow one to analyze road segments with an $M/G/c/c$ Erlang loss queueing model. The equations summarized in Table 2 are a formal description of the "physics" of a single road segment, modelled as an $M/G/c/c$ queue. The equations for $V^{(n)}$ describe the velocity of n vehicles moving along a road segment. Service rate of a road segment can be evaluated with expression for $f(n)$. Equations for P_0 and P_n can be used to compute probability distribution functions of service rates for specific road segments.

5.1. Analytical and simulation model

The $M/G/c/c$ queueing models were implemented as two applications: *LESLAM* and *MGCCSimul*. An application *LESLAM* was developed in DyAMS lab by Smith and Jane, this application numerically solves equations (Table 2) for $M/G/c/c$ state dependent queue (Jain and Smith, 1997). *LESLAM* allows one to evaluate performance parameters of a road link with specific input parameters and arrival rate of vehicles. The second application – the simulation model which is utilized to evaluate a route assignment policy, was developed by Cruz et al. (2005) and implemented as C++ program *MGCCSimul*. The simulation model considers a road segments a_l as $M/G/c/c$ state dependent queue, with Poisson arrival rate λ, general service rate G and limited capacity c. Transportation network with detailed link characteristics, an origin–destination matrix, assigned egress routes and arrival rates for each source are used as an input for this model. The model updates service rates on the road segments dynamically during simulation, applying linear or exponential congestion model for single road segments (Table 2). The $M/G/c/c$ simulation model evaluates total clearance time T_C and total traveled time D_T for specific evacuation plan. In addition, the model provides detail information for every road link a_l of the evacuation network $G_e(N,A)$, such as blocking probability P_C, expected travel time to traverse a road link $E(T_S)$, expected number of evacuees on a road link $E(Q)$ and throughput θ.

5.2. Modeling time delay on road links

Modeling time delays along egress routes is a challenging problem due to the fact that vehicles' velocity depends on number of evacuees in the system, which is a *random* value by definition of arrival process. To get some insight into relation between travel time, properties of a road link and properties of arrival process (arrival rate λ) we performed series of simulations. Using the *MGCCSimul*

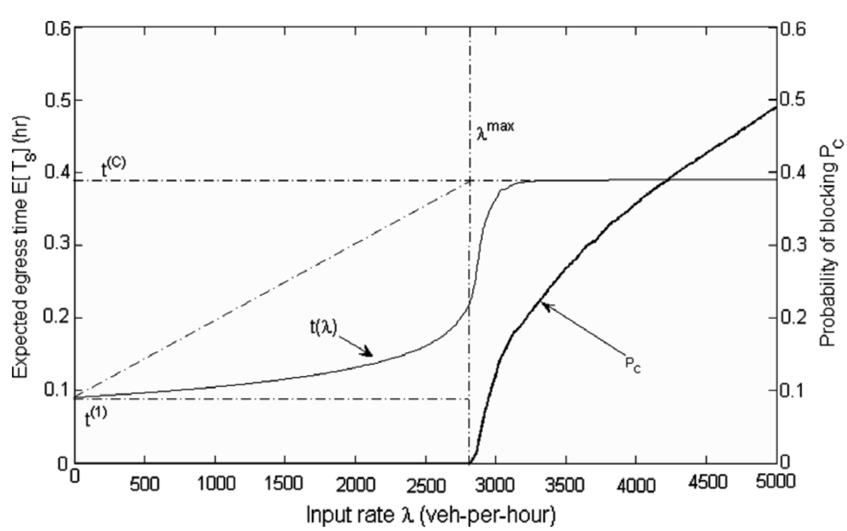

Fig. 6. Expected travel time $E[T_S]$ and probability of blocking P_C as a function of an input rate λ ($L = 5.0$ miles, $W = 1$ lane, $K_{max} = 200$).

program, we evaluated expected time $E[T_S]$ to traverse a road link ($L = 5$ miles, $W = 1$ lane and $K_{\max} = 200$ vehicle/mile/lane) and probability of blocking for a given arrival rate $\lambda \in [0, \ldots, 10{,}000]$. Fig. 6 presents results of simulations. Analysis and comparison of travel time and blocking probability curves (Fig. 6) lead to several conclusions and observations. Expected time t_l to traverse a road link a_l of length L_l and capacity C_l^L is bound by lower (t_l^{lb}) and upper (t_l^{ub}) values (Eqs. (10) and (11))

$$t_l^{lb} = t_l^{(1)} = \frac{L_l}{V_l^{(1)}}, \tag{10}$$

$$t_l^{ub} = t_l^{(C_l^L)} = \frac{L_l}{V_l^{(C_l^L)}}, \tag{11}$$

where $V_l^{(C_l^L)}$ is velocity of vehicles on fully utilized road and can be computed with linear or exponential congestion models (Table 2). Blocking before service ($P_C \geqslant \epsilon$) occurs when an input arrival rate value exceeds some critical value $\lambda_l^{L\max}$. We need to note that a steep increase in travel time occurs in the vicinity of $\lambda_l^{L\max}$ as well. The value of $\lambda_l^{L\max}$ for a road link a_l and specific blocking probability threshold $\epsilon(P_c(C_l^L, \lambda_l^{L\max}) \leqslant \epsilon)$ can be obtained with the $M/G/c/c$ state dependent queueing model and *LESLAM* application. Therefore, we propose (i) to set an upper limit on arrival rate of incoming evacuees to any road link a_l and (ii) to approximate travel time on a road link a_l which is a part of egress routes with a linear function (Fig. 7)

$$t_l(\lambda_l^L) = t_l^{(1)} + \frac{t_l^{(C_l^L)} - t_l^{(1)}}{\lambda_l^{L\max}} \lambda_l^L. \tag{12}$$

The aggregate arrival rate into a link a_l (which composed by arrivals of evacuees traveling different egress routes which share the same road links) can be evaluated as

$$\lambda_l^L = \sum_i \lambda_i \sum_j \sum_k \alpha_{lijk} x_{ijk}. \tag{13}$$

The proposed formulation of travel time delays for stochastic *EEP* assures no blocking during evacuation and prevents over utilization of shortest routes. The travel time function can be approximated with a linear piece-wise approximation as well. We use a linear function to approximate $t_l(\lambda_l^L)$ to not overly complicate the objective function Z (Eq. (3)).

5.3. Description of the algorithm

The proposed approach to solve a stochastic EEP can be summarized with the algorithm below.

Step 1.0 Representation step: Given a regional transportation network $G(N_T, A_T)$, where N_T is a finite set of nodes, A_T is a finite set of edges. Reconfigure the network into a directional one if necessary. Use the k-shortest path algorithm to determine 1st, 2nd, 3rd, …, kth shortest egress routes for each source i (Eppstein, 1998; JimTnez and Marzal, 2003). Using kth shortest path routes, define subsets of road links A and nodes N which comprise evacuation network $G_e(N,A)$, where $A \in A_T$ and $N \in N_T$.

Step 2.0 Analysis step:

Step 2.1 Compute upper bound on arrival rates $\lambda_l^{L\max}$ for each road link $a_l \in A$ to ensure absence of blocking within the evacuation network $G_e(N,A)$.

Step 2.2 Compute maximum travel time $t_l^{(C)}$ for a single road link $a_l \in A$ for all arcs. The value of $t_l^{(C)}$ corresponds to a case when a road link fully utilized. Given $\lambda_l^{L\max}$, $t_l^{(1)}$ and $t_l^{(C)}$ define linear function $t_l(\lambda_l^L)$ for all road arcs a_l.

Step 3.0 Synthesis step

Step 3.1 Formulate and solve the *EEP IP* model according to Eqs. (3) and (4). The solution of the *EEP IP* model will generate a route assignment vector **x**.

Step 3.2 Evaluate the route assignment plan for the evacuation network $G_e(N,A)$ with the *MGCCSimul* program to obtain total clearance time T_C, total traveled time D_T and performance measures for road links A.

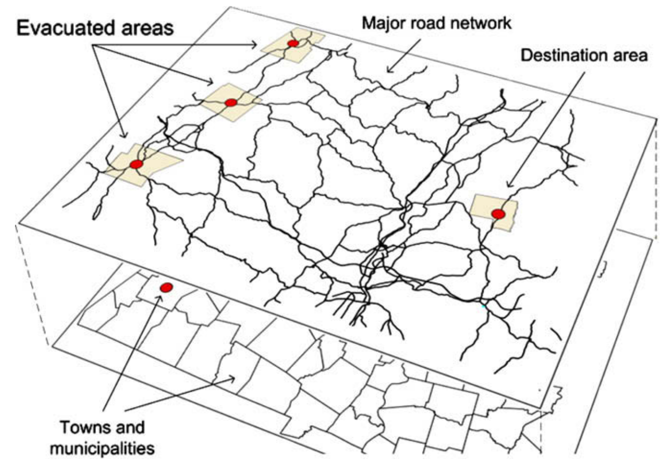

Fig. 8. A regional transportation network: population centers and road infrastructure.

6. Experimental results

In this section, a small case study is utilized to demonstrate the proposed methodology. The approach was tested on the following sample evacuation problem, where the number of evacuees, arrival rates and road network characteristics were chosen to be

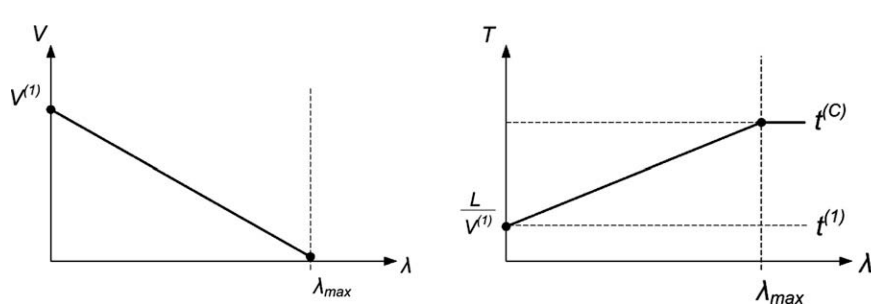

Fig. 7. Simple model of velocity and time as function of arrival rate.

adequate to ensure congestion during peak demand and to not obscure the example, though the suggested model do not have an upper limit on number of evacuees or network size. Therefore, it is necessary to evacuate 4500 households from three neighborhoods S_1, S_2 and S_3 to refuge center D_1 (Fig. 9). We assume that each household will use one vehicle and that evacuees depart from *EPZs* and arrive into evacuation network according to Poisson Process with arrival rates $\lambda_1 = \lambda_2 = \lambda_3 = 1200$ vehicle/h. The task is to design *an optimal evacuation plan (EP)* and evaluate the plan's measures of effectiveness, such as clearance time T_C, total travelled distance D_T and probability of blocking P_C. A road network, which topology was prompted by the larger network shown in Fig. 8, is presented on Fig. 9. Physical characteristics of the sample network, such as length L_l, number of lanes W_l, speed limit $V_l^{(1)}$ and maximum density $K_{\max}$ for each road link a_l, are presented in Table 7.

The *k*th shortest paths ($k = 15$) from each source to refuge center were computed for the sample network with Eppstein's algorithm (Eppstein, 1998; JimTnez and Marzal, 2003). These potential egress routes for each source are presented in Table 8.

6.1. Benchmark evacuation policy

To establish a benchmark for an optimal routing policy, we tested the evacuation plan when all evacuees assigned to the 1*st* shortest egress routes (Fig. 10). This policy (which we will denote *SP* policy) is an example of "common sense" solution, when evacuees choose the shortest (and as perceived the fastest) routes to leave the affected areas. We denote it with the following notation:

$$EP_1 = \{R_{S_1,1}, R_{S_2,1}, R_{S_3,1}\},$$

where egress routes are

$$R_{S_1,1} = \{a_1 - a_5 - a_{12} - a_{19} - a_{22} - a_{26}\},$$
$$R_{S_1,1} = \{a_2 - a_5 - a_{12} - a_{19} - a_{22} - a_{26}\},$$
$$R_{S_3,1} = \{a_3 - a_5 - a_{12} - a_{19} - a_{22} - a_{26}\}.$$

The evacuation plan was evaluated with the *M*/*G*/*c*/*c* simulation software, and according to the simulation results the population was evacuated in 2.95 h and traveled 162,000 miles ($T_C^{SP} = 2.95$ h, $D_T^{SP} = 162,000$ miles). Table 3 summarizes road link performance measures for the *SP* policy (Probability of blocking P_C, throughput θ, average number of vehicles on the road segment $E(Q)$, average traverse time T_S).

Table 3
Evaluation of *SP* route assignment policy with *M*/*G*/*c*/*c* simulation

Road link a_l	P_c	Θ (vehicle/h)	$E(q)$ (vehicle)	$E(T_S)$ (h)	$E(T_1)$ (h)
a_1	0.000	507.8	63.35	0.13	0.03
a_2	0.000	507.8	65.05	0.13	0.03
a_3	0.000	507.8	63.69	0.13	0.03
a_5	**0.385**	1523.4	476.68	0.31	0.08
a_{12}	0.000	1523.4	515.62	0.34	0.13
a_{19}	**0.091**	1523.4	179.39	0.12	0.05
a_{22}	0.000	1523.4	388.62	0.26	0.13
a_{26}	0.000	1523.4	459.76	0.30	0.16

There is significant blocking at entering road links a_5 and a_{19}. For example, 38% of evacuees arriving from road segments a_1, a_2 and a_3 were not able to proceed to road segment a_5 (due to its saturation) and have to stop and wait. Blockage occurs at a_{19} because of its small capacity. In particular we need to compare the parameters of the link leading to a_{19} and a_{19} itself (Tables 4 and 7). Link a_{19} has capacity of 600 vehicles and critical arrival rate 2720 vehicle/h. Link *a*12 which leads to a_{19} has a capacity of 1600 vehicles and a critical arrival rate of 2870 vehicle/h. This particular configuration of these two links and difference in their properties may cause 10% blocking at a_{19}. This policy illustrates that routing solutions belong to the *NI*-set, where shortest routes does not ensure the safest evacuation.

6.2. EEP IP policy

The second policy (IP) was designed with the proposed *EEP IP* model. For the given network $G_e(N,A)$, an upper bound on arrival rates $\lambda_l^{L\max}$ were computed for all road links a_l with a program *LESLAM*. *LESLAM* is a program which allows one to numerically compute performance parameters for a single road link a_l using *M*/*G*/*c*/*c* state dependent queueing models (Table 2). After that parameters $t_l^{(1)}$, $t(\lambda_l^{L\max})$, and $t_l^{(C)}$ were derived (Table 4) for each type of road links. These parameters are required to define a linear approximation of time delay cost functions $t_l(\lambda_l^{L})$ for each road link in accordance with Eq.

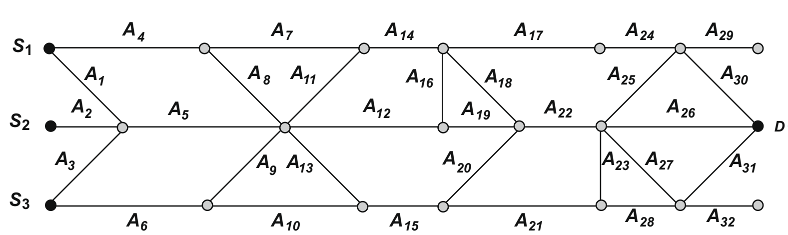

Fig. 9. Sample evacuation network $G_e(N,A)$.

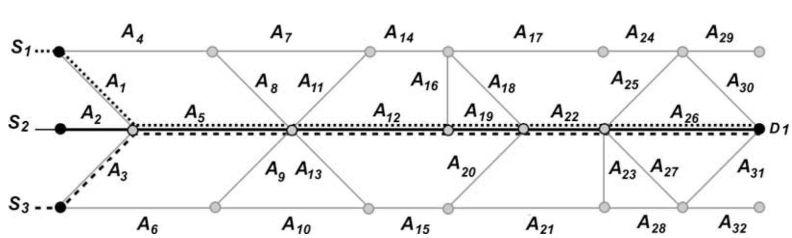

Fig. 10. First shortest path route assignment policy.

Table 4
Road link characteristics

Length L (miles)	Capacity C (vehicle/mile)	$\lambda_l^{L\max}$ (vehicle/h)	$t_l^{(1)}$ (h)	$t(\lambda_l^{L\max})$ (h)	$t_l^{(C)}$ (h)	$P_C^{\max}$	V_l^1 (mile/h)	$V_1(\lambda_l^{L\max})$ (mile/h)	$V_l^{(C)}$ (mile/h)
1.0	200	2600.0	0.02	0.04	0.08	0.630	55.0	27.58	12.94
2.0	400	2680.0	0.04	0.08	0.16	0.635	55.0	26.07	12.85
3.0	600	2720.0	0.06	0.12	0.23	0.638	55.0	25.08	12.82
4.0	800	2750.0	0.07	0.16	0.31	0.634	55.0	24.36	12.81
5.0	1000	2820.0	0.09	0.22	0.39	0.636	55.0	22.46	12.80
6.0	1200	2840.0	0.11	0.27	0.47	0.618	55.0	22.02	12.79
7.0	1400	2850.0	0.13	0.32	0.55	0.621	55.0	22.12	12.79
8.0	1600	2870.0	0.15	0.37	0.63	0.689	55.0	21.67	12.78
10.0	2000	2910.0	0.18	0.47	0.78	0.652	55.0	21.32	12.78
12.0	2400	2980.0	0.22	0.57	0.93	0.625	55.0	20.98	12.88

$W = 1$ and $K_{\max} = 200$ for all road links.

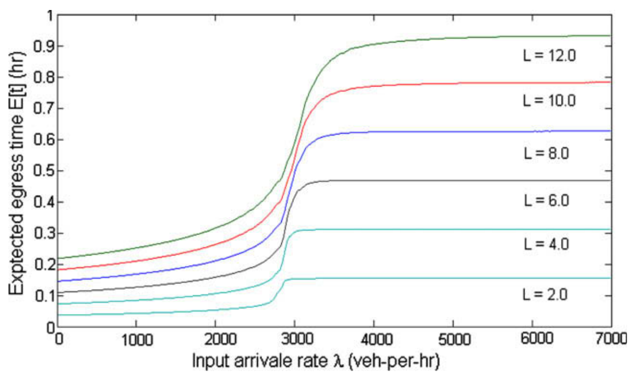

Fig. 11. Family of time delay functions for road links with different length L.

(12). Fig. 11 presents original nonlinear time delay functions. The *EEP IP* model was solved with SAS9.1.3 optimization solver. The optimal decision vector $\mathbf{x}$ for the problem is $\mathbf{x} = \{x_{1,1,1} = 1,\ x_{2,1,10} = 1,\ x_{3,1,11} = 1\}$

Table 5
Evaluation of EEP IP route assignment with $M/G/c/c$ simulation

Road link a_l	P_C	Θ (vehicle/h)	$E(Q)$ (vehicle)	$E(T_S)$ (h)	$E(T_1)$ (h)
a_1	0.000	651.11	29.01	0.05	0.03
a_2	0.000	651.11	29.04	0.05	0.03
a_5	0.000	1302.22	215.68	0.17	0.08
a_6	0.000	651.11	57.84	0.09	0.06
a_9	0.000	651.11	100.49	0.15	0.11
a_{11}	0.000	662.39	102.09	0.15	0.11
a_{12}	0.000	1290.93	327.27	0.25	0.13
a_{14}	0.000	662.40	144.29	0.22	0.16
a_{17}	0.000	662.40	143.72	0.22	0.16
a_{19}	0.000	1290.93	129.36	0.10	0.05
a_{22}	0.000	1290.93	306.03	0.24	0.13
a_{24}	0.000	662.40	72.81	0.11	0.08
a_{26}	0.000	1290.93	364.91	0.28	0.16
a_{30}	0.000	662.40	100.38	0.15	0.11

Table 6
Performance measures for the routing policies

	First SP policy	EEP IP model	Relative change
Evacuation time T_C (h)	2.95	2.30	−22%
Total distance D_T (miles)	162,000	183,260	+13%
Number of blocked links	2	0	
Max. blocking probability	38%	0	−100%

$$R_{S_1,1} = a_1 - a_5 - a_{12} - a_{19} - a_{22} - a_{26},$$
$$R_{S_2,10} = a_2 - a_5 - a_{11} - a_{14} - a_{17} - a_{24} - a_{26},$$
$$R_{S_3,11} = a_6 - a_9 - a_{12} - a_{19} - a_{22} - a_{26}.$$

The *EEP IP* routing policy is to evacuate population from the first source node along the 1st shortest route, the population from source node S_2 should be assigned to the 10th shortest route and 11th shortest route should be used to evacuate population from source node S_3 (Fig. 12). This policy was evaluated with the $M/G/c/c$ simulation program as well.

Table 5 summarizes performance measures for the road links: such as probability of blocking P_C, throughput θ, and average travel

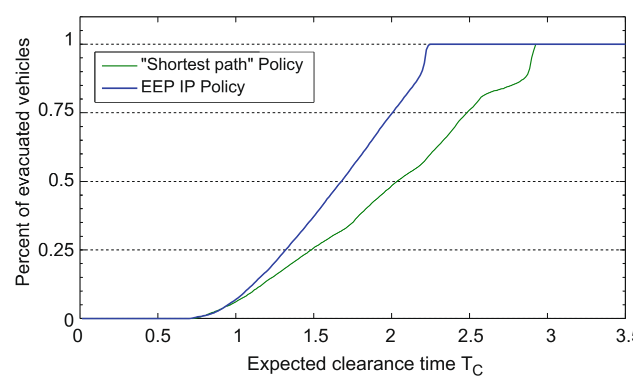

Fig. 13. Clearance time curves for *SP* and *EEP IP* policies.

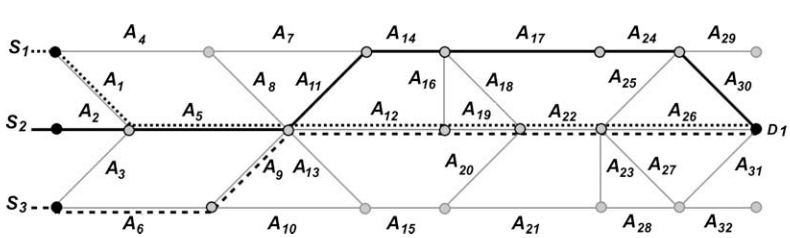

Fig. 12. IP evacuation policy.

Table 7
Sample road network characteristics

Road link a_l	Length (miles)	Road link a_l	Length (miles)	Road link a_l	Length (miles)
a_1	2.0	a_{12}	8.0	a_{23}	1.0
a_2	2.0	a_{13}	7.0	a_{24}	5.0
a_3	2.0	a_{14}	10.0	a_{25}	7.0
a_4	7.0	a_{15}	10.0	a_{26}	10.0
a_5	5.0	a_{16}	1.0	a_{27}	7.0
a_6	4.0	a_{17}	10.0	a_{28}	7.0
a_7	10.0	a_{18}	3.0	a_{29}	3.0
a_8	7.0	a_{19}	3.0	a_{30}	7.0
a_9	7.0	a_{20}	3.0	a_{31}	7.0
a_{10}	10.0	a_{21}	12.0	a_{32}	3.0
a_{11}	7.0	a_{22}	8.0		

$W_l = 1$, $V_l^1 = 55$ miles/h and $K_{\max} = 200$ for all links l

Table 8
First fifteen shortest paths from all origins to destination D_1

S_i	k	Egress route $R_{S_i,1,k}$	Length D_{i1k} (miles)
S_1	1	$a_1 - a_5 - a_{12} - a_{19} - a_{22} - a_{26}$	36
	2	$a_1 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{26}$	37
	3	$a_1 - a_5 - a_{12} - a_{16} - a_{17} - a_{24} - a_{30}$	38
	4	$a_1 - a_5 - a_{12} - a_{19} - a_{22} - a_{25} - a_{30}$	40
	5	$a_1 - a_5 - a_{12} - a_{19} - a_{22} - a_{27} - a_{31}$	40
	6	$a_1 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{25} - a_{30}$	41
	7	$a_1 - a_5 - a_{12} - a_{19} - a_{22} - a_{23} - a_{28} - a_{31}$	41
	8	$a_1 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{27} - a_{31}$	41
	9	$a_1 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{23} - a_{28} - a_{31}$	42
	10	$a_4 - a_8 - a_{12} - a_{19} - a_{22} - a_{26}$	43
	11	$a_4 - a_8 - a_{12} - a_{16} - a_{18} - a_{22} - a_{26}$	44
	12	$a_1 - a_5 - a_{11} - a_{14} - a_{18} - a_{22} - a_{26}$	45
	13	$a_4 - a_8 - a_{12} - a_{16} - a_{17} - a_{24} - a_{30}$	45
	14	$a_1 - a_5 - a_{13} - a_{15} - a_{20} - a_{22} - a_{26}$	45
	15	$a_1 - a_5 - a_{11} - a_{14} - a_{17} - a_{24} - a_{30}$	46
S_2	1	$a_2 - a_5 - a_{12} - a_{19} - a_{22} - a_{26}$	36
	2	$a_2 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{26}$	37
	3	$a_2 - a_5 - a_{12} - a_{16} - a_{17} - a_{24} - a_{30}$	38
	4	$a_2 - a_5 - a_{12} - a_{19} - a_{22} - a_{25} - a_{30}$	40
	5	$a_2 - a_5 - a_{12} - a_{19} - a_{22} - a_{27} - a_{31}$	40
	6	$a_2 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{25} - a_{30}$	41
	7	$a_2 - a_5 - a_{12} - a_{19} - a_{22} - a_{23} - a_{28} - a_{31}$	41
	8	$a_2 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{27} - a_{31}$	41
	9	$a_2 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{23} - a_{28} - a_{31}$	42
	10	$a_2 - a_5 - a_{11} - a_{14} - a_{18} - a_{22} - a_{26}$	45
	11	$a_2 - a_5 - a_{13} - a_{15} - a_{20} - a_{22} - a_{26}$	45
	12	$a_2 - a_5 - a_{11} - a_{14} - a_{17} - a_{24} - a_{30}$	46
	13	$a_2 - a_5 - a_{13} - a_{15} - a_{20} - a_{22} - a_{25} - a_{30}$	49
	14	$a_2 - a_5 - a_{11} - a_{14} - a_{18} - a_{22} - a_{25} - a_{30}$	49
	15	$a_2 - a_5 - a_{13} - a_{15} - a_{20} - a_{22} - a_{27} - a_{31}$	49
S_3	1	$a_3 - a_5 - a_{12} - a_{19} - a_{22} - a_{26}$	36
	2	$a_3 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{26}$	37
	3	$a_3 - a_5 - a_{12} - a_{16} - a_{17} - a_{24} - a_{30}$	38
	4	$a_3 - a_5 - a_{12} - a_{19} - a_{22} - a_{25} - a_{30}$	40
	5	$a_6 - a_9 - a_{12} - a_{19} - a_{22} - a_{26}$	40
	6	$a_3 - a_5 - a_{12} - a_{19} - a_{22} - a_{27} - a_{31}$	40
	7	$a_6 - a_9 - a_{12} - a_{16} - a_{18} - a_{22} - a_{26}$	41
	8	$a_3 - a_5 - a_{12} - a_{19} - a_{22} - a_{23} - a_{28} - a_{31}$	41
	9	$a_3 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{25} - a_{30}$	41
	10	$a_3 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{27} - a_{31}$	41
	11	$a_6 - a_9 - a_{12} - a_{16} - a_{17} - a_{24} - a_{30}$	42
	12	$a_3 - a_5 - a_{12} - a_{16} - a_{18} - a_{22} - a_{23} - a_{28} - a_{31}$	42
	13	$a_6 - a_9 - a_{12} - a_{19} - a_{22} - a_{25} - a_{30}$	44
	14	$a_6 - a_9 - a_{12} - a_{19} - a_{22} - a_{27} - a_{31}$	44
	15	$a_6 - a_9 - a_{12} - a_{16} - a_{18} - a_{22} - a_{25} - a_{30}$	45

time $E(T_S)$. For purposes of comparison, Table 5 contains $E(T_1)$, which is traversal time along a link for an ideal case if all vehicles move up to speed limit. Though expected travel time $E(T_S)$ along road links a_5, a_{12}, a_{19} and a_{22} is almost twice greater than $E(T_1)$, this routing policy copes with the congestion level ($P_C = 0$ for all road links a_l). Moreover, this *IP* routing solution provides evacuation time $T_C^{IP} = 2.30$ h and total travelled distance $D_T^{IP} = 183,260$ and no blocking at any road link due to more balanced road utilization by evacuees. This *EEP IP* evacuation plan can be considered as an optimal one, as it achieved *no blocking* and *minimum clearance time*, therefore it will ensure safe evacuation process.

6.3. Analysis of experiment

Table 6 compares two approaches. Both analyzed evacuation policies are Pareto optimal. Though the policy derived with EEP IP model leads to greater total travelled distance, it simultaneously minimizes blocking and congestion level, as well as clearance time. In comparison with the original *SP* policy, *EEP IP* policy results in 22% decrease in clearance time. The clearance time curves for both policies presented in Fig. 13. These curves were obtained during evaluation of analyzed policies with simulation. The suggested *EEP IP* model provides analytical solution fast and requires only one evaluation with simulation program.

7. Summary and conclusions

In this paper, we have presented a methodology for designing optimal routing policies for emergency evacuation planning (*EEP*). An integer program (*IP*) model formulation was presented to generate an optimal route assignment for a stochastic *EEP*. This *IP* model utilizes state-dependent decaying service rate *M/G/c/c* queueing models to capture time delays functions on road links. The performance measures for the generated evacuation policy, such as clearance time, total traveled distance and congestion level, are evaluated with the *MGCCSimul* simulation software. Such a combination of the analytical optimization model and simulation technique allows decision makers to cope effectively with massive regional evacuation, stochastic nature of evacuees' departure process of and traffic congestion. Finally, the suggested methodology was illustrated with a computational example.

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